

Chun-Fu Chen

M.S. in Applied Mechanics
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SUMMARY

Computational Materials Scientist with an M.S. in Applied Mechanics. Specialized in leveraging First-Principles calculations (DFT), Molecular Dynamics (MD), and Machine Learning Interatomic Potentials (MLIP) to decode complex atomistic mechanisms. Proficient in establishing cross-scale computational frameworks to bridge quantum-level interactions and macroscopic thermomechanical behaviors, seeking to drive innovation in advanced semiconductor modeling and simulation.

EDUCATION

Degree	Institute	Major	Focus	Year
M.S.	National Taiwan University (NTU)	Applied Mechanics	Computational Materials	2024-2026
B.S.	Chung Yuan Christian University	Mechanical Engineering	Solid Mechanics	2020-2024

RESEARCH & ACADEMIC EXPERIENCE

- National Taiwan University (NTU) / Academia Sinica**
Graduate Researcher, Computational Nanomaterial and Nanomechanics Lab

Sep 2024 - Present
Taipei, Taiwan
- Constructed a comprehensive cross-scale simulation framework to decode the micro-level charging mechanisms and thermodynamics of advanced anode 2D carbon-based materials (e.g., 6F-HBC).
 - Engineered DFT calculations using **VASP** to generate high-quality training data sets for Machine Learning Interatomic Potentials (MLIP).
 - Implemented **DeepMD** (MLIP) to bridge quantum-level accuracy with classical MD scale, enabling precise simulation of large-scale structural evolution beyond traditional computational limits.
 - Employed Grand Canonical Monte Carlo (**GCMC**) methods to simulate the continuous multi-site filling process, accurately evaluating the theoretical maximum capacity and volume expansion rate.
 - Automated multiple simulation tasks and streamlined workflows within a **LINUX** HPC environment to optimize computational efficiency.
- Chung Yuan Christian University (CYCU)**
Undergraduate Researcher, Department of Mechanical Engineering

Sep 2020 - Jun 2024
Taoyuan, Taiwan
- Analyzed macroscopic structural deformation, thermal stress, and strain accumulation using principles of **Solid Mechanics** and **Continuum Mechanics**.

TECHNICAL SKILLS

- Modeling & Simulation:** VASP (DFT), LAMMPS (MD), GCMC, DeepMD (MLIP)
- Programming Languages:** Python (Data parsing, Visualization)

- **Operating Systems:** LINUX (HPC environments), Windows
- **Domain Knowledge:** Cross-scale Framework Design, Solid Mechanics, Continuum Mechanics, Computational Chemistry, Thermodynamics

KEY COURSES TAKEN

- **Core Mechanics & Physics:** Advanced Solid Mechanics, Thermodynamics, Dynamics, Quantum Mechanics Principles, Statistical Mechanics
- **Computational & Applied:** Computational Materials Science, Numerical Methods, Molecular Dynamics Simulation, Scientific Computing with Python

SELECTED PROJECTS

- **Investigation of Nano Graphene as Negative Electrode for Sodium-Ion Battery Storage Performance Using Machine Learning Potential and GCMC Method** *2025 - Present*
Master's Thesis Research
 - Developed a unified cross-scale methodology integrating DFT, MLIP, and GCMC to systematically investigate the microscopic sodium storage mechanisms of nano-graphene anode materials.
 - Determined the specific sequential order of adsorption sites based on energy states and successfully predicted the theoretical maximum capacity, providing critical guidance for future anode material design.